



Killara
HIGH SCHOOL

STUDENT NUMBER

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MATHEMATICS

COURSE

ASSESSMENT TASK NO 2
2011

TIME ALLOWED: *Reading time: 5 minutes*
Working time: 3 hours

INSTRUCTIONS: This paper consists of 10 questions
ANSWER all 10 questions
Start each question in a new booklet
Marks may not be given for untidy and poorly arranged work.
All relevant working should be shown.

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

OUTCOMES ASSESSED:

- H1 - Seeks to apply mathematical techniques to problems in a wide range of practical contexts
- H2 - Constructs arguments to prove and justify results
- H4 - Expresses practical problems in mathematical terms based on simple given models
- H5 - Applies appropriate techniques from the study of calculus and trigonometry to solve problems
- H6 - Uses the derivative to determine the features of the graph of a function
- H7 - Uses the features of a graph to deduce information about the derivative
- H8 - Uses techniques of integration to calculate areas and volumes
- H9 - Communicates using mathematical language, notation, diagrams and graphs

This assessment task constitutes 25 % of the Higher School Certificate Course Assessment.

Total marks – 120

Attempt Questions 1 – 10

All questions are of equal value

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Marks

Question 1 (12 marks) Use a SEPARATE writing Booklet.

- (a) Evaluate $\sqrt{\frac{3.37^3 - 2.13}{14.67 \times 10^3}}$ write your answer in scientific correct to 3 significant figures. 2
- (b) Solve for x $1 - 3x \leq 7$ 2
- (c) Solve $x + \frac{2}{x} = \frac{9}{2}$ 2
- (d) Evaluate $3^{2x} \times 9^{1-x}$ 2
- (e) Rationalise the denominator $\frac{2}{2\sqrt{5}+3}$ 2
- (f) Simplify, write your answer with positive index $\left(\frac{1}{3a^{-2}}\right)^2$ 2

Question 2 (12 marks) Use a SEPARATE writing Booklet.

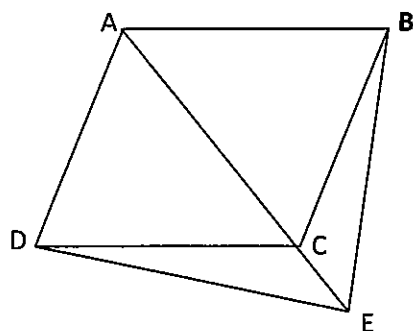
(a) Completely factorise $a^2 - b^2 + a + b$ 2

(b) Solve $|2x - 3| = 1$ 2

(c) Evaluate $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x - 2}$ 1

(d) Solve $a = a^3$ 2

(e) ABCD is a rhombus



NOT TO SCALE

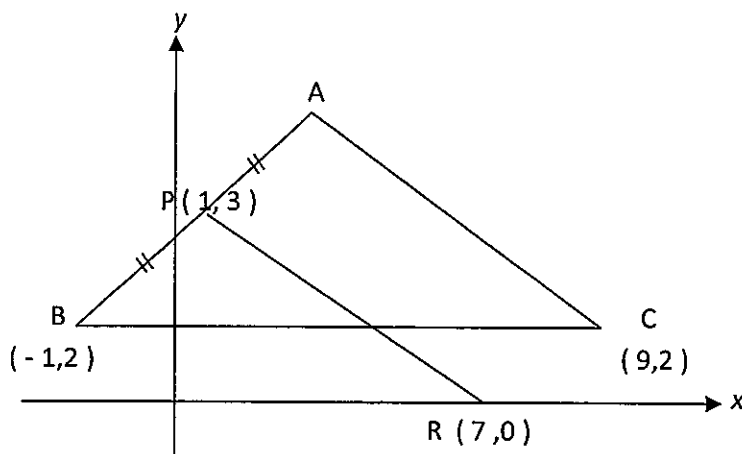
(i) State why $\angle ACD = \angle ACB$ 1

(ii) Prove that $\triangle DCE \equiv \triangle BCE$ 3

(iii) State why $DE = EB$ 1

Question 3 (12 marks) Use a SEPARATE writing Booklet.

- (a) Given that the distance from $A(-1, b)$ to $B(2, 4)$ is $\sqrt{18}$ units. Find one possible value of b . 2
- (b) The points B and C have coordinates $(-1, 2)$ and $(7, 2)$, and P is the midpoint of AB and has coordinates $(1, 3)$.

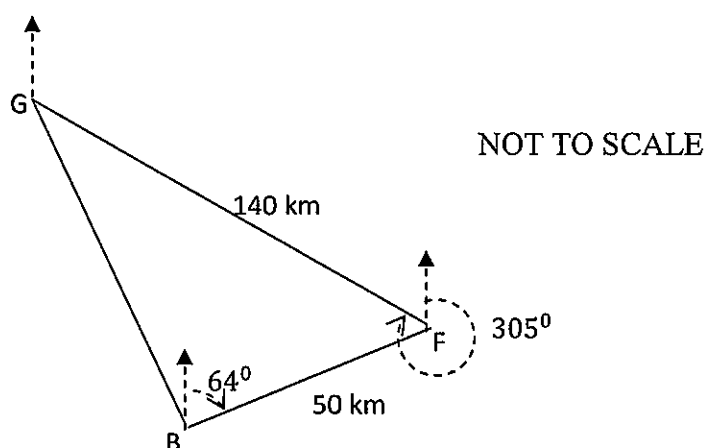


- (i) Show that coordinates of A are $(3, 4)$. 2
- (ii) Write the equation of line BC . 1
- (iii) Show that equation of PR is $x + 2y - 7 = 0$ 2
- (iv) PR cuts y -axis at E . Find the coordinates of E . 1
- (v) Explain how you know AC is not parallel to PR . 2
- (c) l_1 is the line $2x + y + 2 = 0$ and l_2 is the line $3x + ky - 1 = 0$. Find the value of k when l_1 is perpendicular to l_2 . 2

Question 4 (12 marks) Use a SEPARATE writing Booklet.

(a) Write exact value of $\sin 210^\circ$ 1

(b) 'Elwis' the helicopter flies 50 km from Blue Lake 'B' to Forest Hill 'F' on a bearing of 64° , to dump its water and then flies 140 km from Forest Hill 'F' to Gullatown 'G' on a bearing of 305° .



(i) Show that $\angle BFG = 61^\circ$ 2

(ii) Find, to the nearest km, the distance from B to G. 2

(iii) Use the sine rule to find $\angle BGF$ and hence find the bearing of Blue Lake 'B' from Gullatown 'G', correct to the nearest degree. 3

(c) Prove the following identity $1 + \sec^2 \alpha - \tan^2 \alpha = 2$ 2

(d) Solve $\sqrt{3}\cos\theta = \sin\theta$ to find the value of θ , for $0 \leq \theta \leq 2\pi$. 2

Question 5 (12 marks) Use a SEPARATE writing Booklet.

- (a) Factorise $3a^3 - 24$ 2
- (b) Find the values of k for which quadratic function $f(x) = 3x^2 - 4x + (k + 1)$ have 2 real roots. 2
- (c) The roots of the quadratic equation $2x^2 - 4x + 1$ are α and β . 3
Find $\frac{1}{\alpha} + \frac{1}{\beta}$.
- (d) Express $2x^2 + 9x + 4$ in the form : $A(x + 3)^2 + B(x + 3) + C$ 2
- (e) Find the equation of the normal to the parabola $y = 3 - x - x^2$, where it crosses the y – axis. 3

Question 6 (12 marks) Use a SEPARATE writing Booklet.

(a) Differentiate with respect to x .

(i) $4x^2(3 - 6x)^2$ 2

(ii) $2\sqrt{x} + \frac{1}{x^2}$ 2

(iii) $\frac{x^2}{3x-1}$ 2

(b) Show that $f(x) = (x - 1)^3$ is always increasing. 1

(c) For what values of x , is $y = 3x - x^3$ increasing ? 2

(d) Solve $2\sin^2x - 3\sin x + 1 = 0$ for $0 \leq x \leq 360^\circ$. 3

Question 7 (12 marks) Use a SEPARATE writing Booklet.

- (a) (i) Sketch the function $f(x)$ defined by the rule 2

$$f(x) = \begin{cases} x - 1, & x \leq 1 \\ (x - 1)^2, & x > 1 \end{cases}$$

- (ii) Evaluate $f(2) + f(0)$ 1

- (b) Write the domain of the function $y = \frac{1}{\sqrt{2-x}}$ 1

- (c) For the function $f(x) = x^3 - 3x^2 + 5$,

- (i) Find the coordinates of the stationary points and determine their nature 3

- (ii) Explain why there is a point of inflexion at $x = 1$. 2

- (iii) Sketch a neat graph of the function, indicating all important features. 2

- (iv) What is the minimum value of the function in the domain $1 < x < 3$ 1

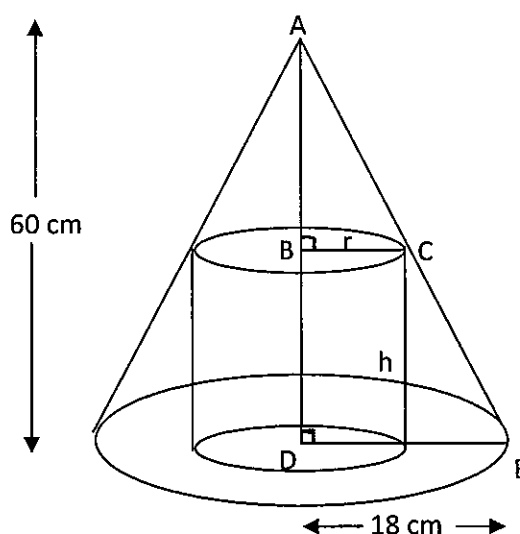
Question 8 (12 marks) Use a SEPARATE writing Booklet.

(a) Find the primitive function of

(i) $(x + 2)(x^2 - 3)$ 2

(ii) $\frac{x^3 - 5x^2 + x}{x}$ 2

(b) A cylinder of height h cm and radius r cm is enclosed in a cone of height 60cm and radius 18 cm.



(i) Prove $\triangle ABC \sim \triangle ADE$ 2

(ii) By using the ratios of the corresponding sides, show that $h = 60 - \frac{10}{3}r$. 2

(iii) Show that the volume of the cylinder is given by $V = 60\pi r^2 - \frac{10}{3}\pi r^3$. 2

(iv) Find the value of r for which the volume of the cylinder is maximum. 2

You must give reasons why your value of r gives the maximum volume.

Question 9 (12 marks) Use a SEPARATE writing Booklet.

(a) Evaluate $\int_0^1 (\sqrt{x} - 3x^2) dx$ 2

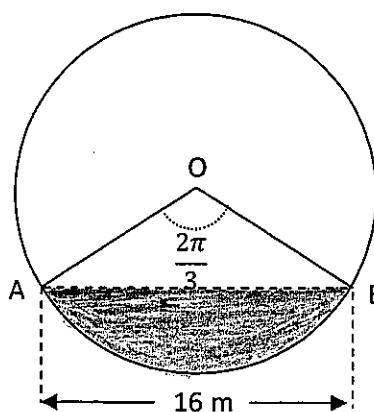
(b) For a certain curve, $\frac{d^2y}{dx^2} = 2x$. The point (3,6) lies on to the curve and its tangent at this point is inclined at 45° to the positive direction of x - axis. Find the equation of the curve. 3

(c) Consider the function $y = \frac{1}{x^2}$. Complete the table below:

x	1	1.5	2	2.5	3	3.5	4
$\frac{1}{x^2}$							

Use Simpson's Rule to approximate $\int_1^4 f(x) dx$. 2

(d) The surface of the water in a horizontal pipe is 16 m wide and subtends an angle of $\frac{2\pi}{3}$ at the centre of the pipe.



Find (i) the distance from the centre of the pipe to the water surface. 2

(ii) the value of the cross sectional area above the water. 3

Question 10 (12 marks) Use a SEPARATE writing Booklet.

- (a) Sketch $y = |x| - 1$ 2
- (b) Find the points of intersection of $y = x^2 + 1$ and $y = x + 3$. 2
- (i) Hence find the area between the two curves. 2
- (ii) Find the volume of the solid generated, when the region in the first quadrant 3
bounded by the curve $y = x^2 + 1$, the line $y = x + 3$ and $y - axis$ is rotated
about $y - axis$.
- (c) Sketch $y = 3\sin 2x - 1$ for $0 \leq x \leq \pi$, clearly marking important features. 3
Write the amplitude and period.

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Mathematics 2011 Half Yearly Assessment

Question 1

a) 4.96×10^{-2}

b) $1 - 3x \leq 7$
 $-3x \leq 6$
 $x \geq -2$

c) $x + \frac{2}{x} = \frac{9}{2}$

$$\frac{2x^2 + 4}{2x} = \frac{9x}{2x}$$

$$2x^2 - 9x + 4 = 0$$

$$x^2 - 8x - x + 4 = 0$$

$$(2x(x-4) - (x-4) = 0$$

$$(2x-1)(x-4) = 0$$

$$x = \frac{1}{2}, 4$$

d) $3^{2x} \times 9^{1-x} = 9^x \times 9^{1-x}$
 $= 9$

OR $3^{2x} \times 3^{2-2x} = 3^2 = 9$

e) $\frac{2}{2\sqrt{3}+3} \times \frac{2\sqrt{3}-3}{2\sqrt{3}-3} = \frac{4\sqrt{3}-6}{20-9}$
 $= \frac{4\sqrt{3}-6}{11}$

f) $\left(\frac{1}{3a^2}\right)^2 = \left(\frac{a^2}{3}\right)^2$
 $= \frac{a^4}{9}$

Award:

2 2 for correct answer
 1 correct rounding incorrect ans

2 2 for correct answer
 1 for $-3x \leq 6$

2 2 for correct answer
 1 for equating the denom
 $2x^2 - 9x + 4 = 0$

2 2 for correct answer
 1 for changing to the same base

2 2 for correct answer
 1 for rationalizing the denominator

2 2 for correct answer
 1 for $\left(\frac{a^2}{3}\right)^2$

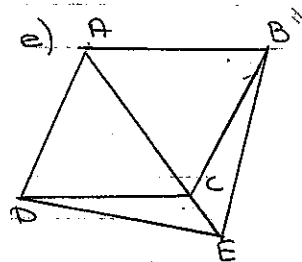
Question 2

a) $a^2 - b^2 + a + b = (a-b)(a+b) + (a+b)$
 $= (a+b)(a-b+1)$

b) $|2x-3|=1$
 $2x-3=1$ $2x-3=-1$
 $2x=4$ $2x=2$
 $x=2$ $x=1$

c) $\lim_{x \rightarrow 2} \frac{x^2+x-6}{x-2} = \lim_{x \rightarrow 2} \frac{(x+3)(x-2)}{(x-2)}$
 $= \lim_{x \rightarrow 2} (x+3)$
 $= 5$

d) $a = a^3$
 $a^3 - a = 0$
 $a(a^2 - 1) = 0$
 $a = 0$ $a^2 = 1$
 $a = \pm 1$



i) $\angle ACD = \angle ACB$ (Diagonal bisect angle at the vertex)

ii) $DC = BC$ (Equal sides of rhombus)

$\angle DCE = 180^\circ - \angle ACD$ (Supplem. angles)

$\angle BCE = 180^\circ - \angle ACB$ (Supplem. angles)

$\therefore \angle DCE = \angle BCE$ (as $\angle ACD = \angle ACB$)

CE common

$\triangle DCE \equiv \triangle BCE$ (SAS)

iii) $DE = BE$ (Corresponding sides of congruent triangles)

Award:

2 2 mark for the correct answer
 1 mark for factorisi
 $a^2 - b^2$

2 1 for each correct solution

1 1 for correct ans

2 2 for 3 solutions
 1 for $a = \pm 1$
 1 for $a = 0, a = 1$
 0 for $a = 1$
 0 for $a = 0$

1 for correct comment

3 3 marks for full solution
 2 for 2 reasons
 (1 for angle + 1 for any other)

1 for correct comment

Question 3

a) $A(-1, b)$, $B(2, 4)$, $d = \sqrt{18}$

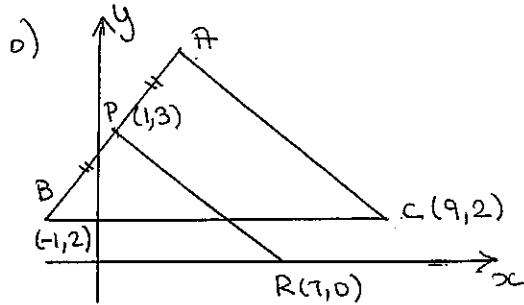
$$18 = (2 - (-1))^2 + (4 - b)^2$$

$$18 = 9 + 16 - 8b + b^2$$

$$b^2 - 8b + 7 = 0$$

$$(b-7)(b-1) = 0$$

$$b = 1 \text{ or } 7$$



i) $\frac{x-1}{2} = 1$ $\frac{y+2}{2} = 3$

$$x-1 = 2$$

$$x = 3$$

$$y+2 = 6$$

$$y = 4$$

$$\therefore A(3, 4)$$

ii) $y = 2$

iii) $P(1, 3)$, $R(7, 0)$

$$m = \frac{3-0}{1-7}$$

$$= -\frac{1}{2}$$

$$y-0 = -\frac{1}{2}(x-7)$$

$$2y = -x + 7$$

$$x + 2y - 7 = 0 \text{ (Equation of PR)}$$

2

Award:

2 for correct answer

1 for the substitution correct

2

1 for each solution

1

1 for $y = 2$

2

2 for the correct equation of the line.

1 for finding gradient and substituting into $y - y_1 = m(x - x_1)$

Question 3 continue

b iv) PR $x + 2y - 7 = 0$

y-intercept when $x = 0$

$$2y = 7$$

$$y = 3\frac{1}{2}, E(0, 3\frac{1}{2})$$

1

Award:

1 for y-coordinate $3\frac{1}{2}$

v) $m_{AC} = \frac{2-4}{9-3}$

$$= \frac{-2}{6}$$

$$= -\frac{1}{3}$$

$$m_{PR} = -\frac{1}{2} \quad m_{PR} \neq m_{AC}$$

$\therefore AC$ is not parallel to PR

2

2 for showing that gradients are not equal

1 Finding gradient of PR

c) $L_1: 2x + y + 2 = 0$ $L_2: 3x + ky - 1 = 0$

$$y = -2x - 2$$

$$m = -2$$

$$ky = -3x + 1$$

$$y = -\frac{3}{k}x + \frac{1}{k}$$

2

2 for finding $k = -6$

1 for finding the gradient of the lines

For perpendicular lines $m_1 \times m_2 = -1$

$$L_1: m_1 = -2$$

$$L_2: m_2 = -\frac{3}{k}$$

$$-2 \times -\frac{3}{k} = -1$$

$$\frac{6}{k} = -1$$

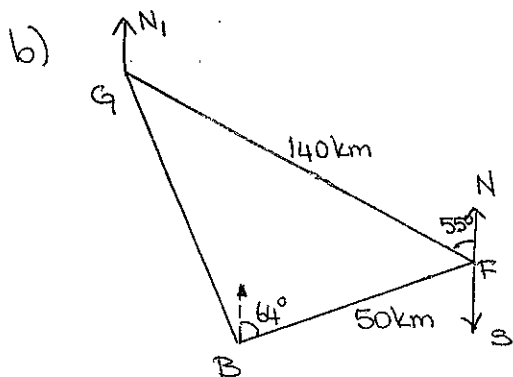
$$k = -6$$

Question 4

$$a) \sin 210^\circ = \sin(180^\circ + 30^\circ)$$

$$\sin 30^\circ = \frac{1}{2}$$

$$\text{3rd Quadrant } \sin 210^\circ = -\frac{1}{2}$$



$$i) \angle BFG = 64^\circ \text{ (Alternate angles)}$$

$$\angle NFG = 360^\circ - 305^\circ \text{ (Angles at a point)}$$

$$= 55^\circ$$

$$\angle GFB = 180^\circ - 64^\circ - 55^\circ \text{ (Supplem. angles)}$$

$$= 61^\circ$$

$$ii) BG^2 = 140^2 + 50^2 - 2 \times 140 \times 50 \times \cos 61^\circ$$

$$= 15312.7$$

$$BG = 123.7 \text{ km} \approx 124 \text{ km}$$

$$iii) \frac{\sin \hat{BGF}}{50} = \frac{\sin 61^\circ}{124 \text{ km}}$$

$$\sin \hat{BGF} = \frac{\sin 61^\circ \times 50}{124}$$

$$= 0.3527$$

$$\angle BGF = 20.7^\circ \text{ (20}^\circ 39')$$

Award

1 for $-\frac{1}{2}$

2 for correct solution for correct

4 for

2 For correct rule and working.
1 for using the rule and correct substitution

3 for correct substitution, angle and bearing
2 Using sin rule and finding the angle in degrees

Question 4 continue

b iii)

$$\angle N_1GF = 180^\circ - 55^\circ \text{ (Co-interior angles)}$$

$$= 125^\circ$$

$$\text{Bearing of B from G: } 125^\circ + 21^\circ$$

$$= 146^\circ \text{ T}$$

$$c) 1 + \sec^2 \alpha - \tan^2 \alpha = 2$$

$$\text{LHS} = 1 + \sec^2 \alpha - \tan^2 \alpha$$

$$= 1 + \frac{1}{\cos^2 \alpha} - \frac{\sin^2 \alpha}{\cos^2 \alpha}$$

$$= \frac{\cos^2 \alpha + (1 - \sin^2 \alpha)}{\cos^2 \alpha} \quad \begin{matrix} (\sin^2 \alpha + \cos^2 \alpha = 1) \\ (\cos^2 \alpha = 1 - \sin^2 \alpha) \end{matrix}$$

$$= \frac{\cos^2 \alpha + \cos^2 \alpha}{\cos^2 \alpha}$$

$$= \frac{2\cos^2 \alpha}{\cos^2 \alpha}$$

$$= 2$$

$$\text{LHS} = \text{RHS}$$

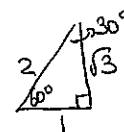
$$\therefore 1 + \sec^2 \alpha - \tan^2 \alpha = 2$$

$$d) \sqrt{3} \cos \theta = \sin \theta$$

$$\sqrt{3} = \frac{\sin \theta}{\cos \theta}$$

$$\tan \theta = \sqrt{3}$$

$$\tan 60^\circ = \sqrt{3} \quad \therefore 60^\circ = \frac{\pi}{3}$$



In the domain $0 \leq \theta < 2\pi$

Tangent is positive in first and third quadr.

$$\therefore 60^\circ, 240^\circ \quad \frac{\pi}{3}, \pi + \frac{\pi}{3} \Rightarrow \theta = \frac{\pi}{3}, \frac{4\pi}{3}$$

Award

1 for using sin rule and substitution

2 for correct working substitution and answer

1 toward half way solving.

2 for finding both angles

1 for finding 1 angle.

Question 5

a) $3a^3 - 24 = 3(a^3 - 8)$
 $= 3(a-2)(a^2 + 2a + 4)$

b) $f(x) = 3x^2 - 4x + (k+1)$
 For 2 distinct real roots $\Delta > 0$
 $\Delta = b^2 - 4ac$
 $= (-4)^2 - 4(3)(k+1)$
 $= 16 - 12k - 12$
 $= 4 - 12k$
 $\Delta > 0 \quad 4 - 12k > 0$
 $-12k > -4$
 $k < \frac{1}{3}$

c) $y = 2x^2 - 4x + 1$
 $\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$
 $= -\frac{-4}{2} \quad = \frac{1}{2}$
 $= 2$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

$$= \frac{2}{\frac{1}{2}}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = 4$$

d) $2x^2 + 9x + 4 \equiv A(x+3)^2 + B(x+3) + C$
 $RHS = A(x^2 + 6x + 9) + Bx + 3B + C$
 $= Ax^2 + 6Ax + 9A + Bx + 3B + C$
 $= Ax^2 + (6A+B)x + 9A+3B+C$

Award

2 2 for full factorising
 1 for $3(a-2)(\dots)$

2 2 for finding correct
 value of k from
 correct working

1 for finding $4 - 12k > 0$

3 3 for finding $\alpha + \beta$,
 $\alpha\beta$ and answer

2 for finding $\alpha + \beta$, $\alpha\beta$
 substitution

1 for finding $\alpha + \beta$, $\alpha\beta$

2 2 for finding A, B, C
 through correct
 working.

1 For for writing *
 (next page)

Question 5 continue

d) continue

Equate the coefficients.

$$2x^2 + 9x + 4 \equiv Ax^2 + (6A+B)x + 9A+3B+C \quad *$$

$$A = 2 \quad (1)$$

$$6A+B=9 \quad (2)$$

$$9A+3B+C=4 \quad (3)$$

sub $A=2$ into (2)

$$6 \times 2 + B = 9 \quad B = -3$$

Sub $A=2, B=-3$ into (3)

$$9 \times 2 + 3 \times (-3) + C = 4$$

$$18 - 9 + C = 4$$

$$C = -5$$

$$\therefore 2x^2 + 9x + 4 = 2(x+3)^2 - 3(x+3) - 5$$

e) $y = 3 - x - x^2$

y-intercept when $x=0 \quad y=3$

$y' = -1 - 2x$ when $x=0 \quad y' = -1$

Gradient of tangent $m_t = -1$

Gradient of normal $m_1 \times m_2 = -1$

$$\therefore m_n = 1$$

Point-gradient formula

$$(0, 3), m=1$$

$$y - 3 = 1(x - 0)$$

$$y - 3 = x$$

$$y = x + 3 \quad \text{Equation of normal}$$

1 mark up to *

3 for finding equation
 of normal through
 correct working

2 finding gradient
 of normal

1 for y-intercept & c

Question 6

a i) $\frac{d}{dx} 4x^2(3-6x)^2 = (3-6x)^2 \cdot 8x + 4x^2 \cdot 2(3-6x) \cdot (-6)$

$$= 8x(3-6x)^2 - 48x^2(3-6x)$$

$$= 8x(3-6x)[(3-6x)-6x]$$

$$= 8x(3-6x)(3-12x)$$

$$= 72x(1-2x)(1-4x)$$

ii) $\frac{d}{dx} (2\sqrt{x} + \frac{1}{2x}) = \frac{d}{dx} (2x^{\frac{1}{2}} + x^{-2})$

$$= 2 \times \frac{1}{2} x^{-\frac{1}{2}} + (-2)x^{-3}$$

$$= \frac{1}{\sqrt{x}} - \frac{2}{x^3}$$

iii) $\frac{d}{dx} \frac{x^2}{3x-1} = \frac{2x(3x-1) - x^2 \cdot 3}{(3x-1)^2} \quad *$

$$= \frac{6x^2 - 2x - 3x^2}{(3x-1)^2}$$

$$= \frac{3x^2 - 2x}{(3x-1)^2}$$

b) $f(x) = (x-1)^3$
 $f'(x) = 3(x-1)^2$
 $f'(x) > 0 \quad \therefore$ curve $(x-1)^3$ is always increasing as $(x-1)^2 > 0$

c) $y = 3x - x^3$
 $y' = 3 - 3x^2$
 $y' > 0 \quad 3 - 3x^2 > 0$
 $3(1-x^2) > 0$
 $1-x^2 > 0$

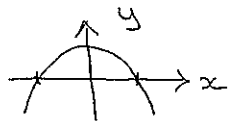
For solution $1-x^2 = 0$

$$x = \pm 1$$

Check $x=0 \quad 1-0^2 > 0$

$$1 > 0 \text{ Yes}$$

$$\therefore -1 < x < 1$$



Award:

2 for full correct differentiation
 1 for first correct line

2 for full correct differentiation

1 for $x^{\frac{1}{2}}$ or x^{-2}

2 for correct differentiation & answer

1 for 1st line *

1 for correct reasoning as $(x-1)^2 > 0$

2 for finding $-1 < x < 1$

1 for $y' > 0$ & $3-3x^2 > 0$

Question 6 continue

a) $2\sin^2 x - 3\sin x + 1 = 0$

Let $a = \sin x$

$$2a^2 - 3a + 1 = 0$$

$$2a^2 - 2a - a + 1 = 0$$

$$2a(a-1) - (a-1) = 0$$

$$(2a-1)(a-1) = 0$$

$$a = \frac{1}{2} \quad a = 1$$

$$\therefore \sin x = \frac{1}{2}, \sin x = 1$$

$\sin$ is positive in 1st & 2nd quadrant

$$\sin x = \frac{1}{2}$$

$$\sin x = 1$$

$$\sin x = \sin 30^\circ$$

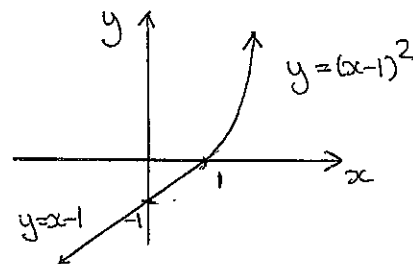
$$\sin 90^\circ = 1$$

$$\therefore x = 30^\circ, 180^\circ - 30^\circ$$

$$\therefore x = 30^\circ, 90^\circ, 150^\circ$$

Question 7

a) $f(x) = \begin{cases} x-1 & x \leq 1 \\ (x-1)^2 & x > 1 \end{cases}$



ii) $f(2) + f(0) = (2-1)^2 + (0-1)$
 $= 1 - 1$
 $= 0$

Award

2 for correct answer

1 for $30^\circ, 90^\circ$

2 For correct drawing of $y = x-1$ & $y = (x-1)^2$ in the correct domain

1 for drawing 1 graph

1 mark for correct answer

Question 7

b) $y = \frac{1}{\sqrt{2-x}}$ $2-x > 0$
 $-x > -2$
 $x < 2$

c) $f(x) = x^3 - 3x^2 + 5$

i) Stationary points when $y' = 0$

$f'(x) = 3x^2 - 6x$

$f'(x) = 0 \quad 3x^2 - 6x = 0$

$3x(x-2) = 0$

$3x = 0 \quad x-2 = 0$

$x = 0 \quad x = 2$

$\therefore x = 0 \quad x = 2 \quad y = 8 - 3 \times 4 + 5$

$y = 5 \quad y = 1$

$\therefore$ Stationary points $(0, 5), (2, 1)$

Nature of stationary points:

$y'' = f''(x) = 6x - 6$

$x = 0 \quad y'' = -6$ concave down maximum pt

$x = 2 \quad y'' = 6$ concave up minimum pt

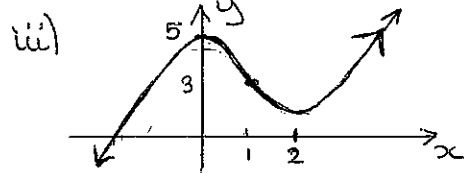
OR a table

ii) $y'' = 0$ when $6x - 6 = 0 \quad x = 1$

Check concavity at $x = 1$

x	0.5	1	1.5
$f''(x)$	-	0	+

Concavity changes $x = 1, y = 3 \therefore (1, 3)$ pt of inflexion



iv) minimum value when $x = 2, y = 3$.

Award

1 for $x < 2$

3 for finding the stationary points (correctly) & determining the nature

2 for finding the stationary points

1 for $3x^2 - 6x = 0$

2 for finding point and determining nature

1 for finding the pt of inflexion

2 for correct drawing and critical points marked

1 for correct drawing

1 for $y = 3$

Question 8

a) i) $\frac{dy}{dx} = (x+2)(x^2-3)$

$= x^3 + 2x^2 - 3x - 6$

$y = \frac{x^4}{4} + 2\frac{x^3}{3} - 3\frac{x^2}{2} - 6x + C$

ii) $\frac{dy}{dx} = \frac{x^3 - 5x^2 + x}{x}$

$= x^2 - 5x + 1$

$y = \frac{x^3}{3} - 5\frac{x^2}{2} + x + C$

b) i

$\angle ABC = \angle ADE = 90^\circ$

Given

$\angle BAC = \angle DAE$ (common)

$\angle ACB = \angle AED$

(Angle sum of a Δ)

$\therefore \Delta ABC \sim \Delta ADE$ (AAA)

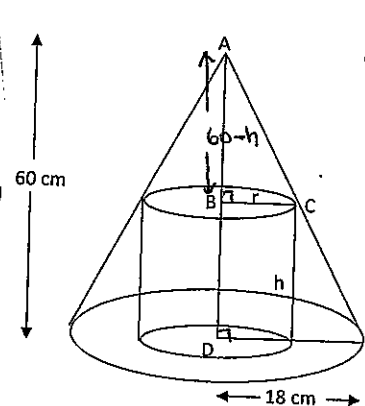
ii) In similar triangles sides are in the same ratio.

$\frac{60-h}{60} = \frac{r}{18}$

$60-h = 60 \frac{r}{18}$

$60 - \frac{10}{3}r = h$

OR $h = 60 - \frac{10}{3}r$



Award

2 for correct answer with "C"

1 for answer without "C"

2 for correct answer

1 for answer without "C"

2 for full proof

2 for full working

1 for writing the ratio

Question 8 continue

b) i) $V = \pi r^2 h$

$$V = \pi r^2 \left[60 - \frac{10}{3}r\right]$$

$$V = 60\pi r^2 - \frac{10}{3}\pi r^3$$

iv) For maximum volume $V' = 0$.

$$\frac{dV}{dr} = 120\pi r - 10\pi r^2$$

$$\frac{dV}{dr} = 0 \quad 120\pi r - 10\pi r^2 = 0$$

$$10\pi r(12 - r) = 0$$

$$r = 0 \quad r = 12$$

$r = 0$ not a solution.

Check the nature of the turning point

r	10	12	13
V'	628.3	0	-408.1

$r = 12$ is a

maximum turning point

$\therefore r = 12$ Volume is maximum

Award

2 2 for correct working
for correct answer

1 Correct substitution

2 2 for finding $r = 12$
determining the
nature.

1 for finding $r = 12$

*
Accurate values of V
for $r = 10, 12, 13$ is a
MUST.

Question 9

a) i) $\int_0^1 (\sqrt{x} - 3x^2) dx = \int_0^1 (x^{\frac{1}{2}} - 3x^2) dx$

$$= \left[\frac{2x^{\frac{3}{2}}}{3} - \frac{3x^3}{3} \right]_0^1$$

$$= \left[\frac{2}{3} \cdot 1 - 1 \right] - 0$$

$$= -\frac{1}{3}$$

2 2 for correct solution
1 for integrating

Question 9 continue

b) $\frac{d^2y}{dx^2} = 2x \quad \frac{dy}{dx} = 2 \frac{x^2}{2} + C$

$$\tan 45^\circ = m = 1$$

$$\therefore 1 = x^2 + C \quad \text{at } (3, 6)$$

$$x = 3 \quad 1 = 9 + C \quad C = -8$$

$$\therefore y' = x^2 - 8$$

$$y = \frac{x^3}{3} - 8x + C_1$$

Sub (3, 6) $6 = \frac{27}{3} - 24 + C_1 \Rightarrow C_1 = 21$

Equation of the curve $y = \frac{x^3}{3} - 8x + 21$

c)

x	1	1.5	2	2.5	3.0	3.5	4
$\frac{1}{x^2}$	1	0.4	0.25	0.16	0.1	0.08	0.0625

or

	1	$\frac{4}{9}$	$\frac{1}{4}$	$\frac{4}{25}$	$\frac{1}{9}$	$\frac{4}{49}$	$\frac{1}{16}$
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$h = 0.5$

$$\int_1^4 \frac{1}{x^2} dx = \frac{0.5}{3} [1 + 4 \times 0.4 + 2 \times 0.25 + 4 \times 0.16 + \dots + 2 \times 0.1 + 4 \times 0.08 + 0.0625]$$

$$= 0.72 \text{ square units}$$

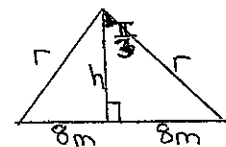
d) i) $\frac{\pi}{3} = 60^\circ$

$$\tan 60^\circ = \frac{8}{h}$$

$$h = \frac{8}{\tan 60^\circ}$$

$$h = 4.62 \text{ m} \leftarrow \left[h = \frac{8}{\sqrt{3}} \text{ m} \right]$$

(evaluate this)



Award

3 3 marks for correct
solution

2 for $y' = x^2 - 8$
finding $m = 1$ & subs
finding

1 for differentiating
 $y' = x^2 + C$

2 2 for correct answer

1 for completing the
table and $h = 0.5$

2 2 Correct solution

1 writing $\tan 60^\circ = \frac{8}{h}$

Question 9 continue

d ii)
 ΔAOB

$$A = \frac{1}{2} ab \sin C$$

$$\sin 60^\circ = \frac{8}{r}$$

$$r = \frac{8}{\sin 60^\circ}$$

$$= 9.2 \text{ m}$$

$$A_{\Delta AOB} = \frac{1}{2} \times 9.2^2 \times \sin 60^\circ$$

$$= 36.95 \text{ m}^2$$

$$\left[\text{OR } A_{\Delta AOB} = \frac{1}{2} \times 16 \times 4.6 \right]$$

$$= 36.8 \text{ m}^2$$

Area of major sector

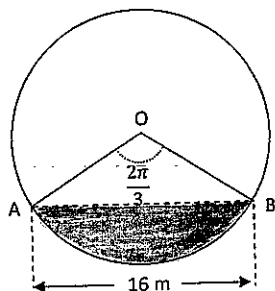
$$A = \frac{\theta r^2}{2} \quad \theta = \frac{4\pi}{3}$$

$$A = \frac{4\pi \times 9.2^2}{3 \times 2}$$

$$= 177.3 \text{ m}^2$$

$$A_{\text{cross section}} = 177.3 + 36.95$$

$$= 214.2 \text{ m}^2$$



3 3 for correct solution
for correct answer

2 for finding area of
 ΔAOB [finding r
and substituting]

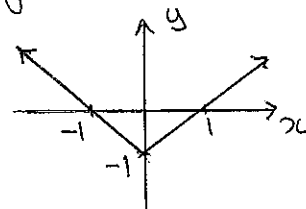
2 for finding $A_{\text{minor sector}}$

1 for finding "r"

1 for finding $A_{\text{minor sector}}$

Question 10

a) $y = |x| - 1$



$$x=0, y=-1$$

$$x=-1, y=0$$

$$x=1, y=0$$

b) $y = x^2 + 1, y = x + 3$

$$x^2 + 1 = x + 3$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

$$x=2$$

$$x=-1$$

$$y=5$$

$$y=2$$

$$\therefore (2, 5) \text{ and } (-1, 2)$$

i)

$$A = \int_{-1}^2 [(x+3) - (x^2+1)] dx$$

$$= \int_{-1}^2 (-x^2 + x + 2) dx$$

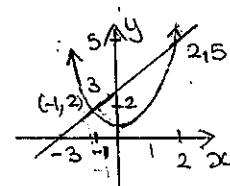
$$= \left[-\frac{x^3}{3} + \frac{x^2}{2} + 2x \right]_{-1}^2$$

$$= \left[-\frac{8}{3} + \frac{4}{2} + 4 \right] - \left[-\frac{-1}{3} + \frac{1}{2} - 2 \right]$$

$$= \frac{20}{6} - \frac{7}{6}$$

$$= \frac{27}{6}$$

$$= 4\frac{1}{2} \text{ square units}$$



Award

2 for correct drawing
and marking axes

1 for basic shape

2 2 for finding
(2, 5) and (-1, 2)

1 for $x^2 - x - 2 = 0$
 $x=2, x=-1$

2 2 for correct integration
and area value

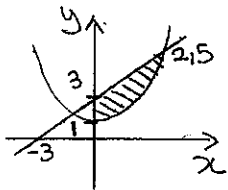
1 for integrating

Question 10 continue

b ii) $y = x^2 + 1 \Rightarrow x^2 = y - 1$

$y = x + 3 \Rightarrow x = y - 3$

$x^2 = y^2 - 6y + 9$



$$V = \pi \int_1^5 (y-1) dy - \pi \int_3^5 (y^2 - 6y + 9) dy$$

$$V = \pi \left[\frac{y^2}{2} - y \right]_1^5 - \pi \left[\frac{y^3}{3} - 6\frac{y^2}{2} + 9y \right]_3^5$$

$$V = \pi \left[\left(\frac{25}{2} - 5 \right) - \left(\frac{1}{2} - 1 \right) \right] - \pi \left[\left(\frac{125}{3} - 3 \times 25 + 45 \right) - \left(\frac{27}{3} - 27 + 27 \right) \right]$$

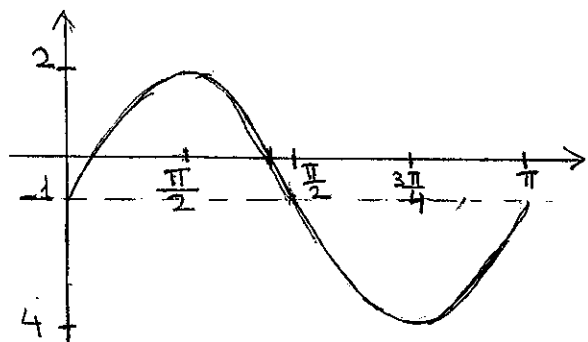
$$V = 8\pi - \pi \frac{8}{3}$$

$$V = \frac{16\pi}{3}$$

$= 16.8$ cubic units

c) $y = 3 \sin 2x - 1$ $0 \leq x \leq \pi$

$A = 3$ period π



Award

3 for correct integration and solution

2 for correct bounds and $\begin{cases} x^2 = y - 1 \\ x^2 = (y - 3)^2 \end{cases}$

and correct substitution

1. for $\begin{cases} x^2 = y - 1 \\ x^2 = (y - 3)^2 \end{cases}$

3 for correct formula and graph

2 for A & P and sketching $3 \sin 2x$

1 for A & P